

Lower temperature tolerance in the temperate, ambush foraging lizard *Pseudocordylus melanotus melanotus*

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Received 5 July 2006; accepted 25 October 2006

Abstract

1. An organism's thermal tolerance to the environment may indicate a thermal limit to its geographical distribution.
2. We measured the lower critical minimum (CTMin) and lower lethal temperature in the lizard *Pseudocordylus melanotus melanotus*.
3. CTMin was 10.17 ± 0.48 °C and the lower lethal temperature was -5.15 ± 0.31 °C.
4. All lizards that froze did not survive. This suggests that these lizards are restricted to areas where the microclimate allows them to avoid freezing.
5. *Pseudocordylus m. melanotus* can allow their body temperatures to drop well below CTMin without any ill effect.

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Keywords: Thermal tolerance; Distribution; Lizard; *Pseudocordylus melanotus melanotus*; CTMin; Lower lethal temperature; Cold exposure

1. Introduction

Temperature has a significant impact on reptilian ecology and distribution (Cowles and Bogert, 1944) since environmental temperature directly influences body temperature (T_b), which in turn, has a significant effect on performance, fitness (Gvoždík and Castilla, 2001) and metabolic expenditure (Dorcas et al., 2004). It is thus not surprising that most lizards attempt to maintain their T_b within a fairly narrow selected range (T_{sel} ; Patterson and Davies, 1978) when conditions allow for it. Generally, the selected T_b range is broadly coincident with optimal temperatures for physiological processes, behaviour and maximum locomotory performance. The upper and lower critical temperatures are measures of the upper and lower thermal tolerances for a species (Doughty, 1994) and these measures can be used to evaluate the thermal requirements and physiology of organisms (Lutterschmidt and Hutch-

ison, 1997a). Critical thermal tolerances also give an indication of the range of climatic conditions that can be tolerated and may be the causal range limiter in some cases (Gaston, 2003; Cossins and Bowler, 1987).

Cowles and Bogert (1944) originally defined the limit to thermal tolerance as the 'thermal point at which locomotory activity becomes disorganised and the animal loses its ability to escape from conditions that will promptly lead to its death'. The end points for the upper and lower critical thermal limits (CTMax and CTMin, respectively) have since been redefined as the temperatures at the upper and lower extreme of tolerance at which the animal cannot right itself when placed on its back, i.e. the loss of righting response (LRR; Doughty, 1994; Lutterschmidt and Hutchison, 1997a). Subsequently, the endpoint for CTMax has again been redefined as the onset of spasms (OS; Lutterschmidt and Hutchison, 1997a). Alexander et al. (1999), however, argue that LRR still satisfies Cowles and Bogert's original definition since locomotor activity is disorganised and the animal is unable to escape. In other ectotherms such as fish, the cessation of spiracle contraction and the loss of equilibrium are sometimes used as endpoints in studies on critical limiting temperatures (Fangue and Bennett, 2003).

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Exposure to critical temperatures need not, in itself, be lethal; only continued exposure, or exposure to temperatures that exceed critical temperatures will lead to death (Hutchison, 1979). Loss of righting is generally considered to be ‘ecologically lethal’, rather than ‘physiologically lethal’ (Hutchison, 1979), because at these temperatures, the animal would be incapable of escaping predators or conditions that could subsequently lead to its death (i.e. temperatures could become more extreme). Although ecological and physiological limits are generally not coincident, it is expected that they are close and covary between species (i.e. species that have low ecologically lethal temperatures should also have low physiologically lethal temperatures). Seasonal and daily patterns of activity are also limited by environmental temperature (Witz, 2001), and, where environmental temperatures are likely to exceed CTMin or CTMax, reptiles are expected to seek thermally buffered shelters.

Many animals rarely encounter extreme temperatures in the wild (Witz, 2001), although there are many documented cases where reptiles do experience and tolerate temperatures that are close to their tolerances. Adaptive tolerance of low temperatures generally involves control or prevention of ice formation (Ramlov, 2000). Some species of reptiles and amphibians regularly experience freezing of body water under natural conditions (termed ‘freeze tolerance’; see Storey and Storey, 1996), whereas other species can tolerate temperatures below freezing, but do not tolerate actual freezing of body fluids (freeze avoidance; e.g. Claussen et al., 1990; Costanzo et al., 1995). Depression of freezing point is usually due to the presence of antifreezes or cryoprotectants (e.g. glycerol and glucose) found in the body fluids, and the extent to which freezing point can be depressed is also usually greater when the animal is in a dry microenvironment.

CTMin, CTMax and lethal tolerances are usually measured under laboratory conditions. Methods for determining thermal tolerances can be grouped into two types: dynamic methods and static methods. CTMin and CTMax are usually measured using a dynamic method, where the animal is cooled or heated at a rate (usually about 1 °C/min) that does not allow thermal acclimation, until LRR, or OS in some cases (Lutterschmidt and Hutchison, 1997a, b). Static methods involve placing the test animal at a single test temperature and the duration to death is measured.

The CTMax and upper lethal limit are usually within a few degrees of each other and normally do not lie much above the voluntary maximum (Heatwole and Firth, 1982). However, because the main effect of low T_b 's is to generally slow reactions, low temperatures are generally less damaging (in terms of biochemical integrity) than high temperatures, which can cause irreversible damage to membranes and proteins (Sherwood et al., 2005). For many species, the upper limit of the range of temperatures that can be tolerated is near the temperature at which proteins begin to denature, which is usually between 45 and

50 °C (Bartholomew, 1977), and this limit generally appears to show little potential for adaptation since it reflects biochemical constraints.

Pseudocordylus melanotus melanotus is a Cordylid lizard that occurs in the cooler regions of southern Africa and can thus be expected to be cold adapted (McConnachie and Alexander, 2004). The species is restricted to rocky outcrops of the South African ‘Highveld’ and Drakensberg escarpment (Branch, 1998) and occurrence is limited to areas where suitable rock crevices are present (crevices must be free of debris and be between 25 and 35 mm wide; McConnachie, 2006). Crevices are used by the lizards as retreat sites during the night and periods of inactivity. Generally, lizards emerge from their crevices from early (summer) to mid-morning (winter) to bask while waiting in ambush for prey (McConnachie and Whiting, 2003), returning to their crevices in the late afternoon. The prevailing climate at our study site, Suikerbosrand Nature Reserve (26°28'40"S; 28°14'12"E) is temperate with summer rainfall. Summers are generally warm and winters are cold, with temperatures regularly dropping below freezing at night. Surprisingly, initial measures of CTMin revealed that some individuals showed LRR at temperatures above 10 °C, significantly higher than temperatures regularly experienced by lizards in the field (McConnachie, 2006). Our aim, therefore, was to establish the lizards' thermal tolerance to critically low temperatures by measuring both the CTMin and the lower lethal temperature for the species under laboratory conditions.

2. Materials and methods

2.1. Lower critical limiting temperature (CTMin)

Our experimental protocol followed that of Lutterschmidt and Hutchison (1997a). Briefly, lower critical limiting temperatures were measured for 10 male and 10 female lizards. Lizards were placed individually in an environmental chamber set at a temperature (−20 °C) so that the lizards cooled at a rate of 1 °C/min. Body temperature was monitored throughout the experiment using a thermocouple inserted approximately 10 mm into the lizard's cloaca and taped to the tail using Micropore™ tape. CTMin was taken as the temperature at which the lizards lost the ability to right themselves once placed on their backs (i.e. LRR). CTMin for males and females was compared using a *t*-test (two-tailed) and the effect of body mass on CTMin was examined using regression analysis.

2.2. Lower lethal temperature

Sixteen lizards (8 females, 8 males) were acclimated at 25 °C for 14 days, in the laboratory. The lower lethal temperature was measured by placing lizards individually in an environmental chamber. Because moisture and partially digested food in the gut may act as ice nucleation sites (see Cossins and Bowler, 1987; Packard et al., 2001), a

small packet of desiccant (Drierite) was placed in the chamber with the lizard and the 14-day acclimation period also allowed for gut clearance. The environmental chamber was connected to a HAAKE C25 water bath with a HAAKE F8 programmable circulator (Fig. 1). The water-bath was filled with alcohol to prevent freezing and it was programmed to cool, until reaching the required minimum temperature, and warm again at rates ($0.03\text{--}0.05\text{ }^{\circ}\text{C}/\text{min}$) similar to those experienced by lizards in the field (McConnachie, 2006). Rates of cooling and warming are important to survival because rapid cooling at subzero temperatures results in the formation of intracellular rather than intercellular ice. In situations where intracellular ice has formed, rapid warming results in less damage than slow warming (Cossins and Bowler, 1987). Slow cooling also enables the full potential for freeze tolerance to be realised, leading to the depression of freezing and supercooling points of intracellular fluids (Cossins and Bowler, 1987).

Each trial lasted 24 h. The water bath was programmed so that the chamber was cooled for 11.5 h until reaching the minimum test temperature, which was maintained for 1 h, and subsequently warmed for another 11.5 h. The start and end temperatures in all cases were $25\text{ }^{\circ}\text{C}$. A single lizard was used for each trial and the minimum temperature of the first trial was $5\text{ }^{\circ}\text{C}$. Thereafter, the minimum temperature of each trial was $5\text{ }^{\circ}\text{C}$ lower until the first death was recorded. Lizards in subsequent trials were cooled to minimum temperatures at $1\text{ }^{\circ}\text{C}$ increments within the $5\text{ }^{\circ}\text{C}$ range above which the first lizard died, i.e. the first four lizards were used as ‘pilot testers’ to determine the range of temperatures to be used in subsequent trials. If lizards did not immediately die from cold exposure during the trial, they were transferred to terraria and allowed an additional 24 h recovery period. All surviving lizards were euthanased after this period. Lizards that died all exhibited an exotherm (a rapid increase in T_b when ice forms). A similar response was reported by Claussen et al. (1990), Churchill and Storey (1992a) and Packard et al. (2001) in amphibians and reptiles, and the exotherm is probably the result of the conversion of latent energy to heat when the

lizard freezes. Thus the T_b at the start of the exotherm was taken as the temperature at which death occurred. The lower lethal temperature was taken as the mean of the temperatures at which the majority of the lizards died (i.e. the mean T_b at the initiation of the exotherm from each lizard). Lizard T_b was monitored using a thermocouple inserted approximately 10 mm into the lizard’s cloaca and taped to the tail. The thermocouple was attached to a MC Systems data logger, which logged temperature every minute over the 24 h test period.

3. Results

3.1. Lower critical limiting temperature

CTMin was $10.17 \pm 0.48\text{ }^{\circ}\text{C}$ (mean $\pm$ SE). CTMin was not affected by sex ($t_{18} = 0.54$, $P = 0.59$; δ : $10.43 \pm 0.67\text{ }^{\circ}\text{C}$, η : $9.90 \pm 0.71\text{ }^{\circ}\text{C}$) and body size had no effect on CTMin (Fig. 2).

3.2. Lower lethal temperature

Of the ‘pilot tester’ lizards, those exposed to 5, 0 and $-5\text{ }^{\circ}\text{C}$ survived. The lizard exposed to $-10\text{ }^{\circ}\text{C}$ died ($n = 4$ ‘pilot testers’). All lizards that died ($n = 9$) at subsequent test temperatures (i.e. between -5 and $-9\text{ }^{\circ}\text{C}$) experienced some degree of freezing. Death was always accompanied by an exotherm spike that occurs when latent heat is released during freezing (Fig. 3). None of the lizards that survived ($n = 3$) showed an exotherm spike.

The point of death was taken as the T_b at which the lizard froze since no lizards that experienced freezing recovered and those that did not freeze all survived. The mean T_b at death was $-5.15 \pm 0.31\text{ }^{\circ}\text{C}$. The range of T_b ’s at which death may occur was -3.38 to $-5.67\text{ }^{\circ}\text{C}$ (Fig. 4).

4. Discussion

The aim of our study was to quantify the lower thermal limits of the lizard *P.m. melanotus* after pilot measurements revealed a surprisingly high CTMin (ca. $10\text{ }^{\circ}\text{C}$) that comfortably exceeded environmental temperatures in the field during winter, when overnight temperatures frequently drop below freezing. *Pseudocordylus m. melanotus* partially offset the lethal effects of low temperatures by supercooling, but were unable to survive temperatures below $-5\text{ }^{\circ}\text{C}$, at which point they experienced freezing. Lillywhite (1987) reports that CTMins for squamates generally range between 1 and $9\text{ }^{\circ}\text{C}$. The CTMin measured for *P. m. melanotus* ($10.17\text{ }^{\circ}\text{C}$) can thus be considered relatively high when compared to other squamates (Table 1), especially for an apparently cold-adapted lizard. There is otherwise little published information available on CTMin in lizards and lower lethal temperatures are rarely investigated. We found a substantial difference (ca. $15\text{ }^{\circ}\text{C}$) between CTMin and lower lethal temperature in *P. m. melanotus* which suggests that, in this species at

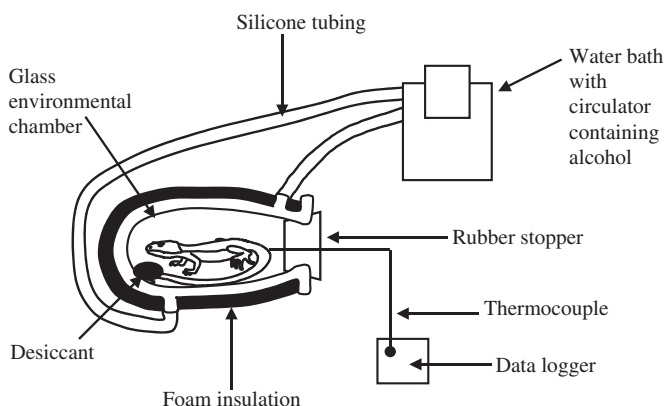


Fig. 1. Diagram showing the environmental chamber and associated equipment used to measure the lower lethal temperature in *Pseudocordylus melanotus melanotus*.

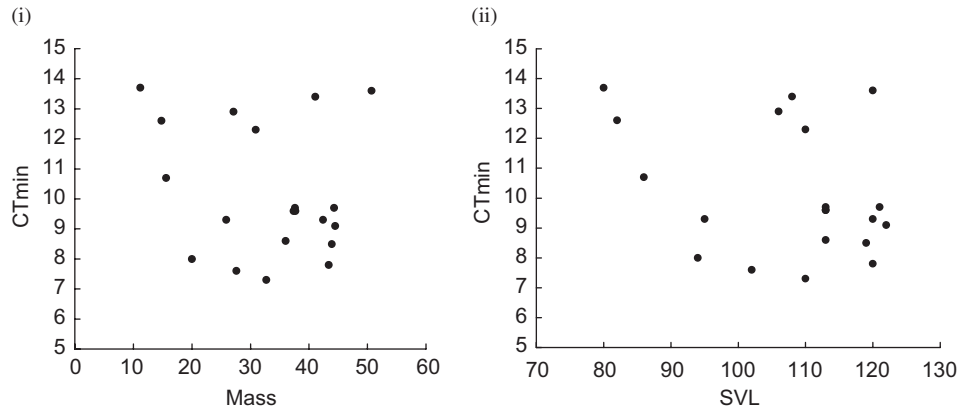


Fig. 2. Relationship between the critical limiting temperature (CTMin, °C) and (i) mass (g); and (ii) snout–vent length (SVL, mm) for all lizards.

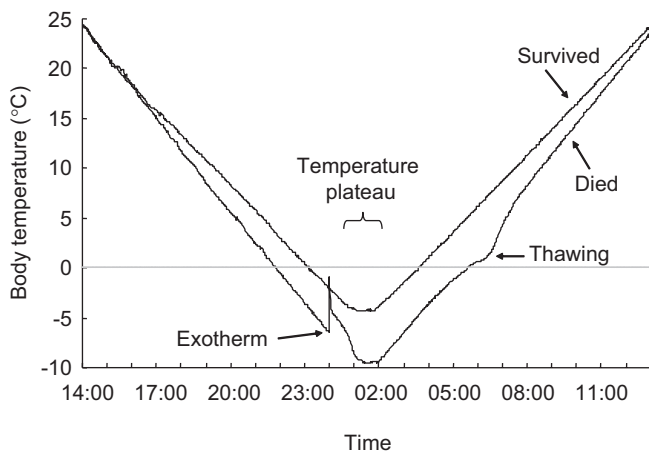


Fig. 3. Lizard T_b for two lizards during cooling experiments over two 24-h periods. Lizards were cooled to test temperature and maintained at that test temperature for 1 h, and then warmed. The upper line shows T_b profile of a lizard that survived, while the lower line shows T_b profile of a lizard that died and exhibited an exotherm spike at the onset of freezing and a lag in increasing T_b near 0 °C when the lizard thaws. The T_b at death was taken as the T_b of the lizard at the start of the exotherm. In this particular case, T_b at death was -6.46 °C.

least, CTMin is not a good measure of the lower thermal limit.

There have been few direct studies on lower lethal limiting temperatures in vertebrates. Most studies deal with cold hardiness or freeze tolerance and usually incorporate some aspect involving survival after exposure to a range of potentially limiting temperatures. Reports of freeze tolerance among squamates are particularly uncommon (Costanzo et al., 1995). Dependence on supercooling is, however, important for species that cannot withstand freezing (Voituron et al., 2002). *Zootoca vivipara* is one lizard species that exhibits both freeze avoidance and freeze tolerance, depending on environmental conditions (Costanzo et al., 1995). The European Wall Lizard (*Podarcis muralis*) can tolerate temperatures as low as -5.9 °C, and can also tolerate freezing at -1.05 °C, but with only 30% survival (Claussen et al., 1990). The percentage of body water that freezes appears to be an important factor in

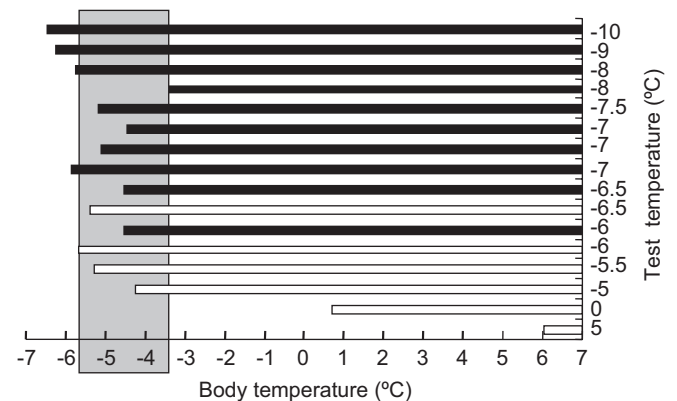


Fig. 4. Body temperature of lizards during lower lethal temperature trials. Black bars indicate lizard T_b at death (i.e. the T_b at the start of the exotherm). Open bars indicate T_b 's of lizards that did not die, and is the minimum T_b experienced by lizards during the trial (not always the same as the minimum chamber temperature due to thermal inertia, etc.). The minimum plateau temperatures for trials are given on the y axis. Shaded area indicates the range of T_b 's at which death might occur and is bounded at the lower edge by the lowest temperature of survival (12th bar from the top), and at the upper edge by the highest temperature at death (4th bar from the top).

freeze tolerance. In *P. muralis*, the maximum ice content tolerated was 28% (Claussen et al., 1990), and *Podarcis sicula* can survive 20% ice content, but not 30% (Burke et al., 2002). The ice content of the frozen *P. m. melanotus* is unknown in this study, but it appears that this species cannot tolerate freezing to any degree since, regardless of the intensity of the exotherm, all lizards that froze, died.

Organisms occurring in temperate, or arctic, climates have developed mechanisms that aid in increasing their supercooling capabilities and reducing the associated stress (Voituron et al., 2002). For animals that avoid freezing, the supercooling point (SCP) is important. *Pseudocordylus m. melanotus* cannot tolerate freezing, but can supercool to approximately -5 °C, and can therefore survive below freezing temperatures, even if they are unable to move or are unconscious (because they would be at T_b 's below their CTMin). The lizards *P. muralis* and *P. sicula* can also

Table 1

Lower critical limiting temperatures (CTMin) reported in lizards and the snake *Hemachatus haemachatus*, which occurs sympatrically with *Pseudocordylus melanotus melanotus*

Species	CTMin (°C)	Reference
<i>Tarentola boettgeri</i>	7.3–9.3	Brown (1996)
<i>Sphenomorphus indicus</i>	♂ 3.9, ♀ 5.9	Ji et al. (1996)
<i>Eumeces chinensis</i>	6.9	Ji et al. (1996)
<i>Eumeces elegans</i>	9.3	Du et al. (2000)
<i>Elgaria multicarinata</i>	9.4 ^a	Kingsbury (1994)
<i>Amphibolurus muricatus</i>	3.0	Heatwole and Firth (1982)
<i>Zootoca vivipara</i>	1.9–2.8	Gvoždík and Castilla (2001)
<i>Hemachatus haemachatus</i>	3.7	Alexander et al. (1999)
<i>Pseudocordylus melanotus melanotus</i>	10.17	This study

^aField active T_b interpreted as CTMin.

supercool and survive exposures to -4 or -5 °C (Claussen et al., 1990; Burke et al., 2002). Invertebrates survive months in a supercooled or frozen state, however, vertebrates may only survive weeks (Voituron et al., 2002). Costanzo et al. (1995) report that *Z. vivipara* can survive supercooling to -3.5 °C for up to several weeks. Torpor may also affect the degree to which an animal can supercool. Nedvĕd (2000) reported that the beetle *Stenotarsus rotundus* can supercool to -7 °C under normal conditions, but can supercool to -13 °C during its diapause stage. *Pseudocordylus m. melanotus* can supercool to approximately -5 °C when acclimated to 25 °C and can potentially tolerate colder temperatures if acclimatized to lower environmental temperatures. It is thus likely that these Crag Lizards can withstand below freezing temperatures for extended periods and be able to survive periods of extreme cold without freezing.

Geographically separated populations may exhibit different cold tolerance abilities, possibly as a result of differing environmental or acclimation temperatures. Kalushkov and Nedvĕd (2000) showed a clear difference in the SCP between two isolated populations of the bug *Pyrrhocoris apterus*, while Spellerberg (1972) found that cold adapted lizards in the genus *Sphenomorphus* had very different tolerance limits to other *Sphenomorphus* that are warm adapted. A within-species investigation of CTMin in warm vs. cold adapted populations of *Trachylepis* (= *Mabuya*) *striata* showed that cold acclimated individuals had a significantly lower CTMin than warm acclimated individuals (Patterson, 1991). (The taxonomy of *Mabuya*, however, has recently been revised (Bauer, 2003), and the two populations in Patterson's (1991) study may be different species.) The *P. m. melanotus* in our study were acclimated at 25 °C for 14 days, which may have resulted in slightly higher than expected CTMin and lower lethal temperature values, but data on T_b 's of field active individuals during the winter months indicate that basking lizards regularly attained T_b 's above 25 °C (McConnachie, 2006). Many studies on thermal tolerances in reptiles neglect to indicate the acclimation temperature or period,

but longer periods are generally used with lower acclimation temperatures where they are reported (periods of between 10 days and 3 months have been reported).

Freeze risk is affected mainly by climate, altitude and latitude, and it may be expected that animals living in cooler regions, or high altitudes or latitudes, may be physiologically adapted to survive lower temperatures (Doughty, 1994). On a smaller scale though, the risk of freezing can also be influenced by local topography, microclimate (e.g. ambient temperature, moisture) and the efficacy of any insulation that may be available (Costanzo et al., 1995). Freeze avoidance can therefore also be achieved by the selection of suitable hibernacula or refugia (Churchill and Storey, 1992a). Rupicolous species such as *P. m. melanotus* can be expected to select crevices that would remain dry and be sufficiently deep to maintain some warmth or buffering from the thermal inertia of the rocks in which they shelter.

The period of exposure to subzero temperatures during our lower lethal temperature measurements in *P. m. melanotus* was less than 12 h (at least half the trial period), and only 1 h at the test temperature. This suggests that cold tolerance in this species is limited to the SCP and lizards can therefore survive environmental temperatures up to approximately -5 °C. If the environmental temperature decreases sufficiently for the lizards to freeze, they will not survive, and a single extreme event could result in large-scale death within a population. It is clear that *P. m. melanotus* will easily survive short periods of subzero temperatures (e.g. overnight frosts) by supercooling, especially if suitable, dry refugia are found and they are in a fasted state. The lizards may also be restricted to areas of summer rainfall since, in areas receiving winter rainfall, refugia may become too moist and ice nucleation may occur as a result of increased exposure to environmental ice.

Acknowledgements

Gauteng Nature Conservation granted permission to collect lizards for this study (permit number 1143). The Animal Ethics Screening Committee of the University of the Witwatersrand cleared all experimental procedures (clearance numbers: 2001/28/2A, 2003/56/6). This study was funded by grants to G.J. Alexander and M.J. Whiting from the National Research Foundation of South Africa and the University of the Witwatersrand. Opinions expressed and conclusions arrived at are not necessarily to be attributed to the NRF.

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